Bayesian Deep Learning for Hyperspectral Image Classification With Low Uncertainty

Xin He, Yushi Chen, *Member, IEEE*, and Lingbo Huang

***Abstract*— In recent years, deep learning models have been widely used for hyperspectral image (HSI) classification, and most of the existing deep-learning-based methods merely focused on high classification accuracy. However, in real applications, classification with low uncertainty matters as much as accurate classification. Unfortunately, the existing methods fail to consider uncertainty. To tackle this challenge, for the first time, Bayesian deep learning (BDL) is investigated to analyze the model uncertainty for HSI classification. Specifically, first, at the feature extraction (FE) stage, an HSI classification framework based on BDL, which contains two Bayesian Gabor layers and a global pooling layer (i.e., BDL-G^2), is proposed. In BDL-G^2, parameters in Gabor layers are sampled from the Gaussian distribution. The proposed BDL-G^2 not only provides the uncertainty estimation but also strengthens the structure characteristic (i.e., texture) of HSI. Second, to model the uncertainty at the final classification stage, BDL-G^2 is combined with a Bayesian fully connected layer (BFL) (i.e., BDL-G^2-BFL), where the parameters' distribution is adjusted adaptively. In the proposed BDL-G^2-BFL, the uncertainty at FE and classification stages is captured, and a whole uncertainty estimation framework is established. Experimental results on the three public HSI datasets demonstrate the superiority in both accuracy and uncertainty. The proposed BDL-based methods pioneer a new direction and provide useful inspiration and experience for practical applications.**

***Index Terms*— Bayesian neural network, deep learning, hyperspectral image (HSI) classification, uncertainty estimation.**

I. Introduction

HYPERSPECTRAL image (HSI) classification tries to allocate a label to each pixel in the scene [1]. Due to the fact that HSI classification is a fundamental technique for HSI processing, for a long time, it is a hot topic in the remote sensing community.

In general, two ingredients, i.e., feature extraction (FE) and following classification, are needed to formulate an HSI classifier. Moreover, the HSI classification performance heavily depends on the effectiveness of FE [2].

In the early stage of HSI classification, many hand-craft-based FE methods have been proposed including spectral and spatial FE methods [3]. For the spectral FE, manifold learning-based methods (i.e., locally linear embedding [4] and Laplacian eigenmap [5]) used the feature embedding in the spectral space for HSI classification. Besides, principal component analysis (PCA) [6] and nonlinear kernel mapping [7] have been used for spectral FE. To increase the classification performance, the spatial features between spatial pixels have been extracted. For example, extended morphological profiles [8] and invariant attribute profiles [9] are the most representative approaches based on morphological operation.

Recently, duo to its powerful FE capability, deep-learning-based methods have been widely used in HSI classification [10]. Deep models hierarchically extract the discriminant and abstract features of HSI and obtain good classification performance [11]. In general, there are many types of deep learning models and most of them have been investigated for HSI FE and classification.

In the beginning of deep-learning-based HSI classification, stacked autoencoder (SAE) and deep belief network (DBN) have been explored [12], [13]. The authors designed proper spectral–spatial deep networks to extract useful features of HSI. However, the SAE and DBN models contain many fully connection layers, which usually have lots of learnable parameters. However, when the number of training samples is limited, lots of parameters usually lead to a problem called overfitting.

Instead of full connections, convolutional neural network (CNN) uses local connections and shared weights to decrease the number of learnable parameters, which obtained good performance in many research areas including HSI classification [14], [15], [16]. For example, a 1-D-CNN with multiple 1-D convolutional kernels was proposed for HSI classification [17]. To add the strong spatial dependencies, Ding et al. [18] explored data-adaptive kernels using 2-D-CNN, where the kernels can be automatically learned from data by clustering. Later, spatial–spectral FEs with 3-D convolutional kernels have explored [19]. In [20], each HSI cube was used as the input of the 3-D-CNN, which caused the heavy computation burden. To reduce the consumption time, in [21], 3-D convolutional kernels were replaced by 2-D convolutional kernels, and residual blocks were investigated by combining 2-D-CNN and 3-D-CNN to extract spatial–spectral features.

Although CNN-based approaches have dominated HSI classification, these methods cannot sufficiently mine the sequence attributes of spectral features. To address this problem, very recently, Transformer has been adopted to HSI classification [22]. In [23], SpectralFormer was proposed by extracting spectrally local sequence features from neighboring bands.

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In addition, combining CNN with Transformer led to better classification performance, and the spectral discrepancies and local spatial context information were captured at the same time [24], [25], [26].

With the help of well-designed architecture, deep-learning-based models have obtained good classification accuracy of HSI. However, the uncertainty estimation of HSI classification is not considered, which refers to the lack of confidence for each sample's output and reflects whether we trust the result. Different from accuracy, which measures whether the predict label is equal to the ground truth, uncertainty offers a probability, assessing the confidence level for each sample's output. In general, there are two types of uncertainty: data uncertainty and model uncertainty [27]. Data uncertainty is caused by inherent property of data distribution such as the presence of noise in the data. While model uncertainty arises from the lack of knowledge about the parameters of the model.

There are only several works focused on the uncertainty in active-learning-based HSI classification. All of them used uncertainty to label samples and alleviate the problem of limited samples in HSI classification [28]. The core idea of these methods is to select the most informative samples using uncertainty estimation. In [29], active learning was first combined with CNNs via the Bayesian theory to perform HSI classification with very limited training sets, which intelligently selected unlabeled training samples using dropout to obtain uncertainty over the model predictions. Later, a new loss function with rank learning was adopted in the uncertainty learning model to select samples [30]. Despite these methods considering the uncertainty in selecting unlabeled samples, they are simply used MC-Dropout without considering the distribution of model weight in the training process [31].

To obtain lower uncertainty and higher accuracy results at the same time in real applications, in this article, Bayesian deep learning (BDL) is investigated to model uncertainty for HSI classification [32]. BDL learns a posterior distribution over the model parameters by observing the training samples, which offers uncertainty estimation via its parameters in form of probability distributions.

At the FE stage, BDL-G^2 is explored for HSI classification, whose key component is the Bayesian Gabor layer. Due to the fact that Gabor filters are effective in extracting textures, the Bayesian Gabor layers are explored in this article to strength the structure information of HSI [33]. Gabor filtering is incorporated into the BDL and the Gabor parameters are sampled by the Gaussian distribution and optimized by variational Bayesian inference [34]. Furthermore, at the classification stage, a model-free distribution with parametric prior in the fully connected (FC) layer is proposed to model uncertainty, whose masks on all the neurons adjust adaptively. Combining BDL-G^2 with a Bayesian FC layer (BFL), BDL-BDL-G^2-BFL is proposed to constitute an HSI classification framework with low uncertainty.

The main contributions of this article can be concluded as follows.

1) A new classification framework, i.e., BDL, is first investigated for HSI classification, which obtains low uncertainty and accurate classification at the same time. BDL infers distribution over the model's weights and thus provides the estimation of model uncertainty.
2) BDL-G^2 is proposed for HSI classification low uncertainty, which uses two Bayesian Gabor layers to model uncertainty at the FE stage. Parameters in the Gabor layers are reparametrized by Gaussian distribution, and each parameter is represented by learned means and variances which is optimized by variational Bayesian inference.
3) Furthermore, a simple and effective BDL-G^2-BFL is proposed for HSI classification, which captures the model uncertainty at both the FE and classification stages. In BDL-G^2-BFL, a model-free distribution is explored by adjusting parametric prior distribution of the FC layer adaptively.

The rest of this article is organized as follows. Sections II and III present the detailed elaboration of the proposed BDL-G^2 and BDL-G^2-BFL, respectively. Section IV conducts the experiments analyses and results. Finally, Section V concludes this article.

II. Proposed BDL-G^2 for HSI Classification

Fig. 1 displays the overall framework of the proposed BDL-G^2, which composes of the FE stage and classification stage. At the FE stage, parameters in Bayesian Gabor layers are sampled from the Gaussian distribution to model uncertainty for HSI classification, and each Gabor parameter is represented by learnable means and variances in Gaussian distribution. After that, the extracted features are fed into ReLU operation, max-pooling layer, and a Global average pooling (GAP) layer. Finally, at the classification stage, an FC layer is used to finish the HSI classification task.

A. Bayesian Gabor Layer for HSI FE

HSI contains abundant spectral and spatial information. Therefore, proper extraction of spectral and spatial features is a crucial step for a successful HSI classifier. In the proposed BDL-G^2, PCA is used to extract the spectral features of HSI, and the two Bayesian Gabor layers are used to extract spatial features.

Similar to the mammalian vision systems, Gabor filtering has been proved an effective technique for extracting the texture representations by descripting of frequency and orientation, which has been widely applied in face recognition, texture-based material classification, and so on [35]. Inspired by Gabor filtering, the structure features of HSI can be enriched by modeling texture information with orientation-sensitive functions [36].

Specifically, a Gabor filter with real and imaginary components can be represented by the following equation:

$$G_{\lambda,\alpha,\phi,\delta,\gamma} = \exp\left\{-\frac{\tilde{E}^2+\gamma^2\tilde{F}^2}{\delta^2}\right\}\exp\left\{i\left(2\pi\frac{\tilde{E}}{\lambda}+\phi\right)\right\} \quad (1)$$

$$\tilde{E} = E\cos\alpha + F\sin\alpha \quad (2)$$

$$\tilde{F} = -E\sin\alpha + F\cos\alpha \quad (3)$$

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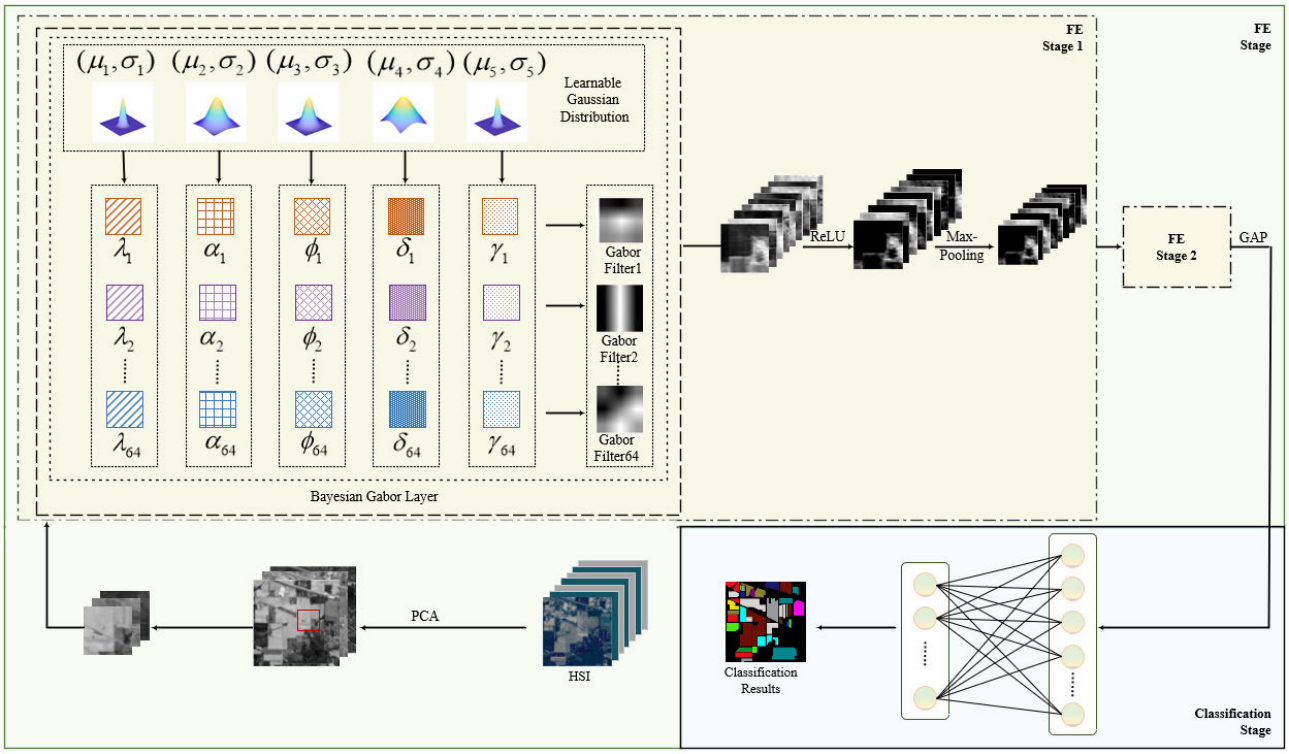


Fig. 1. Overall framework of the proposed BDL-G^2 for HSI classification.

where $G_{\lambda,\alpha,\phi,\delta,\gamma}$ is a Gabor filter. E and F are the coordinates. $\tilde{E}$ and $\tilde{F}$ are the transformed coordinates. λ is the wavelength of the sinusoidal component. δ represents the standard deviation of the Gaussian envelope. α is the normal direction of parallel stripes. ϕ is the phase offset, and γ indicates the spatial aspect ratio.

Due to the fact that the existing deep-learning-based HSI classification methods are failed to provide model uncertainty, in this article, BDL is investigated in this article. BDL is a field at the intersection between deep learning and Bayesian probability theory, which provides uncertainty estimation by leveraging the hierarchical representation power of deep learning and also is able to infer posterior distributions [37]. Typically, BDL generates uncertainty estimation by placing Bernoulli distribution over the network's weights called Monte Carlo dropout (MC-Dropout) sampling.

Instead of simply using the BDL by placing the distribution over the weights for HSI classification, in this article, Gabor filtering is incorporated into BDL to model uncertainty. We take the proposed BDL-G^2 as a probabilistic model [38], whose weights are modeled by the five Gabor parameters (λ, α, ϕ, δ, and γ) and each parameter is represented by Gaussian distribution with learned means and variances.

As shown in Fig. 1, the Gabor parameters are obtained indirectly, and the mean and variance in Gaussian distributions are calculated by the reparameterization [39]. Suppose $\boldsymbol{v}$ is sampled from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma})$, to ensure the standard deviations of the variational distributions are always nonnegative. $\boldsymbol{\sigma}$ is calculated indirectly with $\boldsymbol{\rho}$, which is a learnable parameter and sampled from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma})$

$$\boldsymbol{\theta} = \boldsymbol{\mu} + \boldsymbol{v} \otimes \boldsymbol{\sigma} \tag{4}$$

$$\boldsymbol{\sigma} = \log\left[1 + \exp(\boldsymbol{\rho})\right] \tag{5}$$

where $\otimes$ indicates pointwise multiplication, and $\boldsymbol{\theta}$ represents the posterior distribution of the Gabor parameters, which should be learned in Section II-B.

B. Bayesian Inference of BDL-G^2 for HSI Classification

To obtain model uncertainty for HSI classification, after constructing the Bayesian Gabor layer, the proposed BDL-G^2 should explore Bayesian modeling of the five Gabor parameters.

Let training set $\boldsymbol{D} = \{\boldsymbol{X}, \boldsymbol{Y}\}$ consist of HSI training data $\boldsymbol{X} = \{\boldsymbol{x}_i\}_{i=1}^N$ and the corresponding label set $Y = \{y_i\}_{i=1}^N$ with N training samples. $p(\boldsymbol{W})$ is the prior distribution over these Gabor parameters, and $p(\boldsymbol{W}|\boldsymbol{D})$ is the distribution of Gabor parameters $\boldsymbol{W}$ conditional on the training samples D. $p(\boldsymbol{D}, \boldsymbol{W})$ represents the joint distribution between the five Gabor parameter set $\boldsymbol{W}$ and training samples $\boldsymbol{D}$.

According to Bayes' theorem [40]

$$p(\boldsymbol{W}|\boldsymbol{D}) = \frac{p(\boldsymbol{D}|\boldsymbol{W})p(\boldsymbol{W})}{\int p(\boldsymbol{D}|\boldsymbol{W})p(\boldsymbol{W})d\boldsymbol{W}}. \tag{6}$$

The denominator in the posterior distribution is called the marginal likelihood, or the evidence. $p(\boldsymbol{D}|\boldsymbol{W})$ is the probability which indicates a dataset induced by a prior over parameters $p(\boldsymbol{W})$, which influences the inductive biases of a model. Therefore, the key distinguishing property of a Bayesian approach is to represent solutions given by all the settings of parameters weighted by their posterior probabilities.

To improve the generalization capability, Bayesian model average (BMA) is used. In BDL, an often-overlooked benefit

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of BMA is that BMA could improve accuracy by taking the advantage of Bayesian marginalization, which improves the generalization property of model from a probabilistic perspective [41].

From a probabilistic perspective, generalization depends largely on two properties, the support and the inductive biases of a model [41]. In [41], the support refers to the range of datasets, and the inductive biases indicate the relative prior probabilities of different datasets. Ideally, a model supports a wide range of datasets. However, the model has inductive biases, which provide a high prior probability for a particular class. BMA is defined in [42]

$$p(y|\boldsymbol{x}, \boldsymbol{D}) = \int p(y|\boldsymbol{x}, \boldsymbol{W}) p(\boldsymbol{W}|\boldsymbol{D}) d\boldsymbol{W}. \tag{7}$$

The outputs are y indexed by inputs $\boldsymbol{x}$. Rather than betting everything on one hypothesis with a single setting of parameters $\boldsymbol{W}$, which is weighted by their posterior probabilities, BMA uses all the settings of parameters. This procedure is called marginalization of the parameters $\boldsymbol{W}$. By averaging the predictions of many high-performing models that disagree in some cases, classification accuracy is improved.

However, the posterior over Gabor parameters can represent many complementary solutions to a given problem [43]. Moreover, sophisticated posterior $p(\boldsymbol{W}|\boldsymbol{D})$ is difficult to compute due to the high dimension of $p(\boldsymbol{W}|\boldsymbol{D})$.

To address the closed form and high dimension in the BDL-G^2, variational inference is introduced to find the Gabor parameters $\boldsymbol{W}$ of the distribution on another estimated weights $q(\boldsymbol{W}|\boldsymbol{\theta})$ [44]. Variational inference aims to minimize the Kullback–Leibler (KL) divergence with the true Bayesian posterior on the weights

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{D}) = \arg\min_{\theta} \mathrm{KL}[q(\boldsymbol{W}|\boldsymbol{\theta})||P(\boldsymbol{W}|\boldsymbol{D})]. \tag{8}$$

Therefore, (8) can be referred to the sum of the likelihood cost and a prior-dependent part, computing the sophisticated posterior $p(\boldsymbol{W}|\boldsymbol{D})$ becomes an optimization problem in variational inference. This cost function only needs to consider the relationship between the HSI training data $\boldsymbol{D}$ and the simplicity prior of Gabor parameters $p(\boldsymbol{W})$.

In the proposed BDL-G^2, the loss is computed as follows:

$$\mathcal{L}_1 \approx \sum_{b=1}^{B} [\lambda_b(\log q(\boldsymbol{W}|\boldsymbol{\theta}) - \log p(\boldsymbol{W})) - \log p(\boldsymbol{D_b}|\boldsymbol{W})] \tag{9}$$

where $\boldsymbol{D_b}$ represents the bth mini-batch in HSI data, and $\log q(\boldsymbol{W}|\boldsymbol{\theta}) - \log p(\boldsymbol{W})$ is the KL loss between $q(\boldsymbol{W}|\boldsymbol{\theta})$ and $p(\boldsymbol{W})$. $\log p(\boldsymbol{D_b}|\boldsymbol{W})$ indicates the negative log-likelihood loss. $\lambda_b = 1/N$ is the weight which aims to adjust weight between the KL loss and negative log-likelihood loss.

Instead of simply applying the Gaussian distribution with zero mean and one variance, the prior distribution $p(\boldsymbol{W})$ is defined a mixture of two normal distribution. Therefore, the prior over the Gabor parameters can be defined by the following equation:

$$p(\boldsymbol{W}) = \prod_j \left[0.5\mathcal{N}\left(\boldsymbol{W_j}|0, \boldsymbol{\sigma_1}^2\right) + 0.5\mathcal{N}\left(\boldsymbol{W_j}|0, \boldsymbol{\sigma_2}^2\right)\right] \tag{10}$$

where $\boldsymbol{W_j}$ denote the jth Gabor parameters in the proposed BDL-G^2. $\mathcal{N}$ is the Gaussian distribution.

Inspired by the variational inference, the Gabor parameters are learned by the variational distributions $q(\boldsymbol{W}|\boldsymbol{\theta})$, which avoids the sophisticated posterior computations. Therefore, the output of the proposed BDL-G^2 could reflect the uncertainty by computing model's output probabilities for HSI classification.

III. Proposed BDL-G^2-BFL for HSI Classification

In the proposed BDL-G^2-BFL, two types of model uncertainty at FE and classification stages are captured. The unified architecture of the proposed BDL-G^2-BFL is displayed in Fig. 2. Based on the proposed BDL-G^2, a Bayesian FC layer is investigated to model HSI classification uncertainty, and the parameters in prior distribution of Bayesian FC layer are adjusted adaptively.

A. Model-Free Distribution for HSI Classification Stage

Usually, deep-learning-based HSI classification consists of two stages, including an FE stage and a classification stage, and the parameters in both the stages are single-point. However, since there are many parameters in the model, it might be uncertain which model parameters to choose to obtain the final classification results.

To address this problem, based on the BDL, this article establishes HSI classification with low uncertainty by putting probability distributions over weights to estimate the model uncertainty. In the last section, the proposed BDL-G^2 estimates model uncertainty at the FE stage; however, it is unjustifiable to use single-point estimates as weights at the HSI classification stage. Thus, a Bayesian FC layer is presented to further estimate model uncertainty in HSI classification.

Inspired by the MC-Dropout, a Bayesian FC layer is designed in this article to model uncertainty for HSI classification. MC-Dropout is an effective technique to capture model uncertainty, which casts neurons of the model by multiplying Bernoulli distributed masks as approximate Bayesian inference [31]. MC-Dropout stochastically samples from the network with different random dropout masks. The statistics of this distribution of outputs will reflect the model uncertainty.

However, the drop rate is assigned with fixed value. To address this problem, a model-free distribution is investigated for HSI classification, whose drop rate is adjusted adaptively by choosing the masks randomly. The detailed formation of model-free distribution is described as follows.

Suppose the parameter set in the Bayesian FC layer is defined by $\boldsymbol{\zeta}$. The sampling rule is computed by the following equation:

$$\boldsymbol{O} = \boldsymbol{m} \odot (\boldsymbol{\zeta} \times \boldsymbol{I}) \tag{11}$$

where $\boldsymbol{I}$ and $\boldsymbol{O}$ indicate the input and output of the Bayesian FC layer, respectively. $\boldsymbol{m}$ is the mask vector. $\odot$ is the Hadamard product operation.

In the proposed BDL-G^2-BFL, a model-free distribution $\boldsymbol{g}$ is explored, which learns adaptive parameters in an end-to-end manner to adjust drop rate adaptively. Specifically, in the

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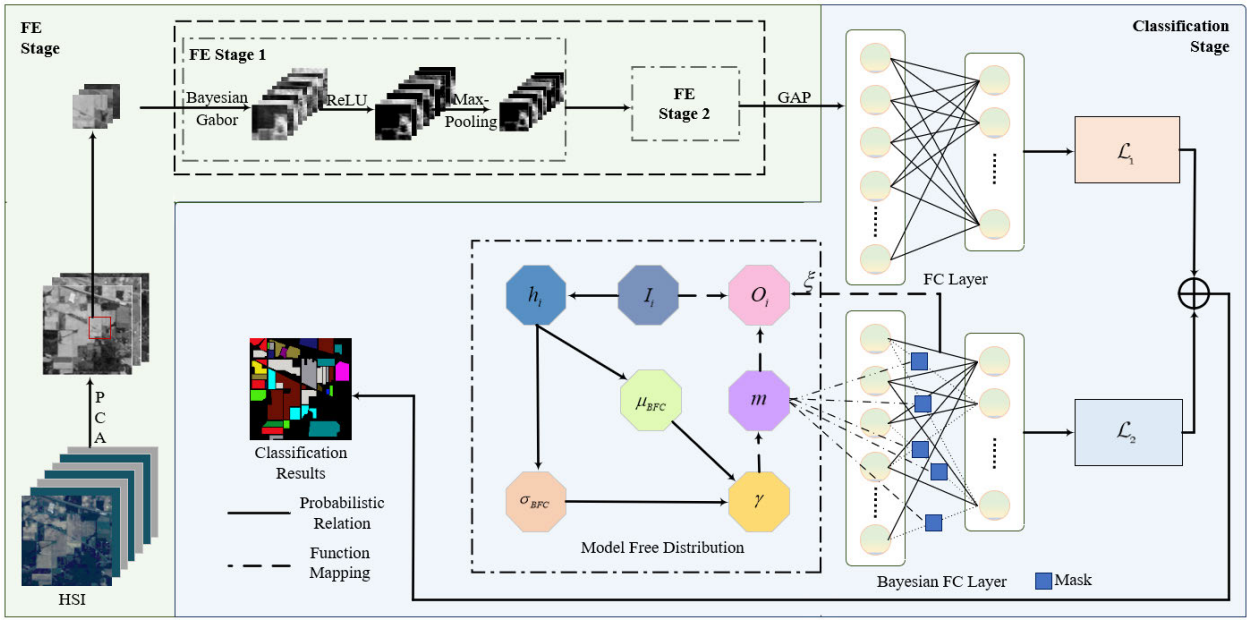


Fig. 2. Unified architecture of the proposed BDL-G^2-BFL for HSI classification.

Bayesian FC layer, the jth element of the mask satisfies $\boldsymbol{m}_j \sim \boldsymbol{g}(\boldsymbol{m}_j)$. A new hidden seed variable $\boldsymbol{r}_j$ is defined following a seed distribution $\boldsymbol{s}(\cdot)$. To obtain the [0, 1]-bounded continuous variable $\boldsymbol{m}_j$, a mapping function $\boldsymbol{k}(\cdot)$ (i.e., Sigmoid function) is introduced which is monotonic and differentiable. The value space of $\boldsymbol{k}(\cdot)$ is in the interval of [0, 1]. Thus, $\boldsymbol{m}_j = \boldsymbol{k}(\boldsymbol{r}_j)$.

The relationship between $\boldsymbol{s}(\cdot)$, $\boldsymbol{k}(\cdot)$, and $\boldsymbol{g}$ is calculated as follows, where $\boldsymbol{k}^{-1}(\cdot)$ is the inverse function of $\boldsymbol{k}(\cdot)$. Note that $\boldsymbol{k}(\cdot)$ is a monotonic function, and therefore, $\boldsymbol{k}^{-1}(\cdot)$ can always be obtained

$$\boldsymbol{g}(\boldsymbol{m}_j) = \boldsymbol{s}\left(\boldsymbol{k}^{-1}(\boldsymbol{m}_j)\right)\frac{d\boldsymbol{k}^{-1}(\boldsymbol{m}_j)}{d\boldsymbol{m}_j}. \tag{12}$$

Once the above conditions are met, the model-free distribution $\boldsymbol{g}$ can be modeled by the dropping mask $\boldsymbol{m}$.

B. Prior Distribution for Parameters in Bayesian FC Layer

After designing the model-free distribution form of $\boldsymbol{g}$, to obtain the model uncertainty at the FC layer for HSI classification, the prior distribution parameters (i.e., $\boldsymbol{\mu}_{\text{BFL}}$ and $\boldsymbol{\sigma}_{\text{BFL}}$) are required.

The prior of $\boldsymbol{\mu}_{\text{BFL}}$ and $\boldsymbol{\sigma}_{\text{BFL}}$ by integrating these features lead the proposed model-free distribution to an optimal distribution. The probability of these masks can be adjusted by optimizing distribution parameters with the prior, and the prior distribution $p(\boldsymbol{\mu}_{\text{BFL}}, \boldsymbol{\sigma}_{\text{BFL}}|\boldsymbol{I})$ is defined by the following equation:

$$p(\boldsymbol{\mu}_{\text{BFL}}, \boldsymbol{\sigma}_{\text{BFL}}|\boldsymbol{I}) = \int p_{\mu_{\text{BFL}}}(\boldsymbol{\mu}_{\text{BFL}}|\boldsymbol{h})\, p_{\sigma_{\text{BFL}}}(\boldsymbol{\sigma}|\boldsymbol{h})\, p_h(\boldsymbol{h}|\boldsymbol{I})\, d\boldsymbol{h} \tag{13}$$

where $\boldsymbol{h}$ are the multivariate Gaussian distributed hidden states.

Suppose $\boldsymbol{\mu}_{\text{BFL}}$ and $\boldsymbol{\sigma}_{\text{BFL}}$ follow Gaussian and inverse gamma distributions (i.e., $\boldsymbol{Ig}$), the prior distribution of $\boldsymbol{\mu}_{\text{BFL}}$ and $\boldsymbol{\sigma}_{\text{BFL}}$ is defined as follows:

$$\boldsymbol{\mu}_{\text{BFL}} \sim \mathcal{N}(\boldsymbol{\mu}_{\text{BFL}}|\cdot) \approx p_{\mu_{\text{BFL}}}(\boldsymbol{\mu}_{\text{BFL}}|\boldsymbol{h}) \tag{14}$$

$$\boldsymbol{\sigma}_{\text{BFL}} \sim \boldsymbol{Ig}(\boldsymbol{\sigma}_{\text{BFL}}|\cdot) \approx p_{\sigma_{\text{BFL}}}(\boldsymbol{\sigma}_{\text{BFL}}|\boldsymbol{h}) \tag{15}$$

$$\boldsymbol{h} \sim \mathcal{N}(\boldsymbol{h}|\cdot) \approx p_h(\boldsymbol{h}|\boldsymbol{I}). \tag{16}$$

To obtain the distribution $p_{\mu_{\text{BFL}}}(\boldsymbol{\mu}_{\text{BFL}}|\boldsymbol{h})$ and $p_{\sigma_{\text{BFL}}}(\boldsymbol{\sigma}_{\text{BFL}}|\boldsymbol{h})$, multilayer perceptrons (MLPs) with Gaussian outputs are designed to approximated $\boldsymbol{\mu}_{\text{BFL}}$ and $\boldsymbol{\sigma}_{\text{BFL}}$. Here, suppose $\boldsymbol{\gamma} = \{\boldsymbol{\Omega}_{\mu_{\text{BFL}}}, \boldsymbol{\Omega}_{\sigma_{\text{BFL}}}, \boldsymbol{b}_{\mu_{\text{BFL}}}, \boldsymbol{b}_{\sigma_{\text{BFL}}}\}$ represents the weights and the bias of the MLPs, and these weights and bias are optimized by the cross entroy loss named $\mathcal{L}_2$.

Therefore, the parameter matrix $\boldsymbol{f}(\boldsymbol{\varepsilon}, \boldsymbol{\zeta}, \boldsymbol{\lambda})$ of drop masks in the Bayesian FC layer is computed as follows:

$$\begin{aligned}\boldsymbol{f}(\boldsymbol{\varepsilon}; \boldsymbol{\zeta}, \boldsymbol{\lambda}) &= \text{diag}(\boldsymbol{m}) \times \boldsymbol{\zeta} \\ &= \text{diag}(\text{Sigmoid}(\boldsymbol{\mu}_{\text{BFL}} + \boldsymbol{\sigma}_{\text{BFL}} \times \boldsymbol{\varepsilon})) \times \boldsymbol{\zeta}\end{aligned} \tag{17}$$

where $\boldsymbol{\varepsilon}_j \sim \mathcal{N}(0, 1)$, $j = 1, \ldots, K$. $\boldsymbol{\varepsilon}_j$ is sampled from the Gaussian distribution with zero mean and unit variance. $\boldsymbol{\lambda} = \{\boldsymbol{\mu}_{\text{BFL}}, \boldsymbol{\sigma}_{\text{BFL}}, \boldsymbol{\gamma}, \boldsymbol{\delta}\}$.

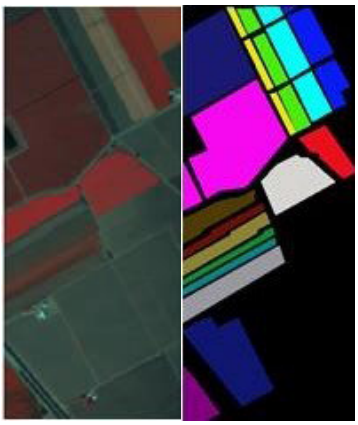
After constructing the generated model-free distribution $\boldsymbol{g}(\boldsymbol{m}|\boldsymbol{\mu}_{\text{BFL}}, \boldsymbol{\sigma}_{\text{BFL}})$ and the parametric prior $p(\boldsymbol{\mu}_{\text{BFL}}, \boldsymbol{\sigma}_{\text{BFL}}|\boldsymbol{n})$ indirectly, as shown in Fig. 2, the proposed BDL-G^2-BFL combined with BDL-G^2 is optimized by the sum of $\mathcal{L}_1$ and $\mathcal{L}_2$.

IV. Results

A. Experimental Settings

1) Datasets Description: The Salinas dataset was acquired by the airborne visible infrared imaging spectrometer (AVIRIS) over Salinas Valley, CA, USA, in 1998. It includes 512 × 217 pixels with 3.7-m spatial resolution. There are 224 spectral bands in the range of 400–2500 nm. After removing 20 water absorption bands (108–112, 154–167, 224),

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(a) (b)

Fig. 3. Salinas dataset. (a) False-color composite image and (b) ground-truth map.

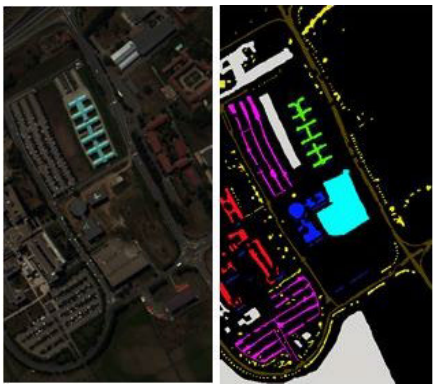
TABLE I
SALINAS LABELED SAMPLE COUNTS

Class			Sample
No.	Color	Name	Numbers
1		Brocoli-green-weeds-1	1977
2		Brocoli-green-weeds-2	3726
3		Fallow	1976
4		Fallow-rough-plow	1394
5		Fallow-smooth	2678
6		Stubble	3959
7		Celery	3579
8		Grapes-untrained	11213
9		Soil-vinyard-develop	6197
10		Corn-senesced-green-weeds	3249
11		Lettuce-romaine-4wk	1058
12		Lettuce-romaine-5wk	1908
13		Lettuce-romaine-6wk	909
14		Lettuce-romaine-7wk	1061
15		Vinyard-untrained	7164
16		Vinyard-vertical-trellis	1737
Total			53785

204 bands were used for analysis. In all, 16 classes are labeled in the reference dataset. The false-color composite of this dataset and the corresponding ground truth are shown in Fig. 3. Table I gives the sample numbers for each class.

The Pavia dataset was recorded by the reflective optics system imaging spectrometer (ROSIS) sensor over the University of Pavia, Northern Italy. A region of 610 × 340 pixels with 1.3-m spatial resolution is used. There were 115 bands whose spectral range is from 430 to 860 nm, and 12 noisy bands were removed. The remaining 103 spectral channels are used in the experiments. There are nine land-cover types in the available ground truth. Its ground-truth map and detailed information regarding the distribution of samples per class are shown in Fig. 4 and Table II.

The Indian Pines dataset was acquired by the AVIRIS sensor over agricultural areas in North Western Indiana. The spatial size of this dataset is 145 × 145 with 20-m spatial resolution. There were 200 bands by discarding 20 water absorbing bands.

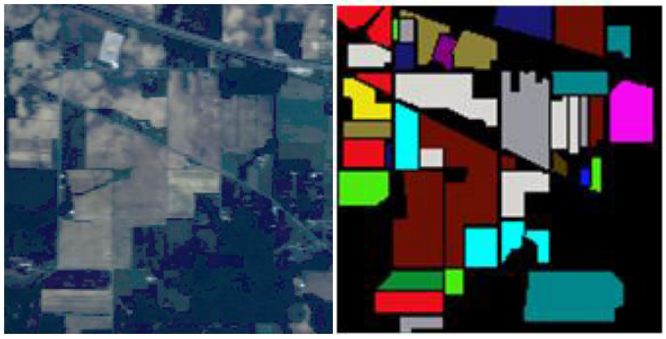


(a) (b)

Fig. 4. Pavia dataset. (a) False-color composite image and (b) ground-truth map.

TABLE II
PAVIA LABELED SAMPLE COUNTS

Class			Sample
No.	Color	Name	Numbers
1		Asphalt	6851
2		Meadows	18686
3		Gravel	2207
4		Trees	3436
5		Metal sheets	1378
6		Bare soil	5104
7		Bitumen	1356
8		Bricks	3878
9		Shadow	1026
Total			43922



(a) (b)

Fig. 5. Indian Pines dataset. (a) False-color composite image and (b) ground-truth map.

The wavelengths of these spectral bands ranged from 400 to 2500 nm. The pseudocolor and the corresponding ground truth map with 16 land-cover categories are shown in Fig. 5. Table III reports the number of samples for each class.

2) Experimental Setup: In the experiments, the training samples are randomly chosen from all the labeled samples in the HSI datasets. The total number of training samples is set to 200 to train the network. And the spatial size of the input HSI sample is set to 33 × 33 to fully extract the spectral–spatial information. In the training process, the batch size is set to 32 for the proposed BDL-G^2 and BDL-G^2-BFL. The learning

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TABLE III
INDIAN PINES LABELED SAMPLE COUNTS

Class			Sample Numbers
No.	Color	Name	
1		Alfalfa	46
2		Corn-notill	1428
3		Corn-min	830
4		Corn	237
5		Grass-pasture	483
6		Grass-trees	730
7		Grass-pasture-mowed	28
8		Hay-windrowed	478
9		Oats	20
10		Soybean-notill	972
11		Soybean-mintill	2455
12		Soybean-clean	593
13		Wheat	205
14		Woods	1265
15		Buildings-Grass-Trees	386
16		Stone-Steel-Towers	93
Total			10249

rate is given with 0.0015, 0.01, and 0.01 for the proposed BDL-G^2 on the Salinas, Pavia, and Indian Pines datasets, respectively. For the proposed BDL-G^2-BFL, the leaning rate is given as 0.01 for all the datasets. The number of epochs is set to 350 on the three datasets for the proposed BDL-G^2 and BDL-G^2-BFL. To alleviate the overfitting problem, PCA is adopted to reduce the trainable parameters in the proposed methods (i.e., BDL-G^2 and BDL-G^2-BFL). In the network, there are five parameters in the Gabor filters, including the wavelength of the sinusoidal component $\boldsymbol{\lambda}$, standard deviation of the Gaussian envelope $\boldsymbol{\delta}$, normal direction of parallel stripes $\boldsymbol{\alpha}$, phase offset $\boldsymbol{\phi}$, and spatial aspect ratio $\boldsymbol{\gamma}$. And all the Gabor parameters are sampled from the Gaussian distribution instead of fixed values.

3) Evaluation Indices: To evaluate the performance of all the proposed methods for HSI classification, four indices including the overall accuracy (OA), average accuracy (AA), Kappa coefficient (K), and expected calibration error (ECE) are adopted as quantitative evaluations. Specially, ECE measures the confidence level of different methods.

Specifically, all the samples need to be grouped into interval bins according to the predicted probability, and ECE computes a weighted average of this error across bins [45]

$$\mathrm{ECE} = \sum_{\mathrm{bin}=1}^{\mathrm{Bin}} \frac{n_{\mathrm{bin}}}{T} |\mathrm{acc}(\mathrm{bin}) - \mathrm{conf}(\mathrm{bin})| \tag{18}$$

where n_{bin} and T represent the number of predictions in each bin and the total number of samples, respectively. Bin is the total number of intervals. The acc(bin) and conf(bin) indicate the accuracy and confidence in each bin, respectively. The smaller the value of ECE leads to the lower the uncertainty of the model.

B. Experimental Results of BDL-G^2

1) Comparisons With Different Methods: The proposed BDL-G^2 is compared with five classical classification approaches, including support vector machine (SVM) [46], spectral–spatial residual network (SSRN) [21], CNN [47], modified MLP [48], PCA-CNN, and MC-Dropout [49]. Specifically, CNN and PCA-CNN consist of three various kernel sizes including 5×5, 4×4, and 3×3 to extract discriminative features. Specially, to validate the effectiveness of the proposed methods, MC-Dropout was adopted in [31], which conducted a theoretical treatment of a network having dropouts in its layers. To estimate the uncertainty of a model, MC-Dropout keeps the dropout layers active during the test and draws multiple output samples for each input. In practice, this is equivalent to performing stochastic forward passes through the model and it obtains multiple outputs for each sample, and one can use the distribution of the outputs of the different passes to estimate the mean and variance of the distribution. The mean is then used for computing classification results, and the variance as a measure of uncertainty.

The classification results of the proposed BDL-G^2 are reported at the penultimate column in Tables IV–VI. As can be seen, the accuracy of the proposed BDL-G^2 reaches 91.97%, 91.31%, and 87.74% in terms of OA on the three datasets. Take the Indian Pines dataset as an example, the proposed BDL-G^2 has the gains of 7.99%, 5.37%, 1.58%, and 0.56% over SVM, SSRN, CNN, and MLP, respectively. Besides, the value of ECE in the proposed BDL-G^2 is the lowest compared with other approaches. The results demonstrate that the proposed BDL-G^2 is an effective technique, which improves the accuracy and reduces the model uncertainty.

2) Distribution Analysis of Output Probabilities: In this experiment, to illustrate that the proposed BDL-G^2 is able to reduce the model uncertainty, the histogram is used in this section to represent the distribution of the output probabilities for all the methods, which are shown in Figs. 6–8 on the Salinas, Pavia, and Indian Pines datasets, respectively. The horizontal axis and the vertical axis represent the mean predicted probability and the distribution of output probabilities, respectively.

It can be observed that most of the test samples are distributed from 0.8 to 1. Take the Salinas dataset as an example, the output probabilities of the proposed BDL-G^2 are distributed around 1 for almost all the samples, which indicates the BDL-G^2's output probabilities for all the samples have high confidence. While for the MC-dropout, most of the test samples are located within the interval of 0.9–1. However, the number of these test samples with high output probabilities is less than the proposed BDL-G^2, which indicates that BDL-G^2 is an effective technique for reducing uncertainty.

3) Visualization of Gabor Filters: To validate the Gabor filters in the proposed BDL-G^2 is able to learn different orientations, Figs. 9 and 10 show several learned Gabor filters in the first and second Bayesian Gabor layers on the Indian Pines dataset, respectively. The original size of the Gabor filter is 3×3; to give a better visualization, the original size is interpolated to 50×50 by the nearest interpolation technique. As can be seen, the Gabor filters have various orientations. For examples, Figs. 9(a) and 10(a) indicate the horizontal direction, Figs. 9(b) and 10(b) represent the vertical direction, and Figs. 9(d)–(h) and 10(d)–(h) display different tilt directions.

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TABLE IV
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE SALINAS DATASET USING 200 TRAINING SAMPLES

Method	SVM	SSRN	CNN	MLP	PCA-CNN	MC-Dropout	**BDL-G^2**	**BGN-G^2 -BFL**
OA(%)	83.09±1.08	90.81±2.87	89.79±8.30	91.92±2.45	91.02±2.27	91.08±4.76	91.97±3.62	92.73±1.36
AA(%)	85.46±2.06	90.77±3.27	90.10±4.25	92.05±3.41	86.09±4.23	92.15±3.76	90.72±4.41	92.39±2.93
K×100	81.07±1.19	89.89±3.25	88.70±9.02	90.96±2.74	89.90±2.57	90.01±5.34	90.96±4.13	91.84±1.54
ECE (%)	29.39±4.32	21.59±5.64	13.61±6.72	5.75±0.78	17.33±2.14	12.13±2.98	5.83±3.24	3.93±1.54
Brocoli_green_weeds_1	94.15±0.50	47.59±63.61	66.96±38.74	87.56±17.77	68.30±25.43	96.31±7.18	94.48±8.34	96.92±2.99
Brocoli_green_weeds_2	98.57±0.89	99.99±0.02	99.47±0.85	91.52±4.04	95.31±3.09	96.88±5.87	95.18±5.04	95.61±4.84
Fallow	90.56±0.50	65.27±9.61	96.05±4.79	87.03±15.21	84.03±28.68	90.27±18.71	88.50±10.05	94.46±4.86
Fallow_rough_plow	98.93±0.40	99.19±0.63	98.06±2.40	97.15 ±3.94	90.74±11.60	98.88±1.31	91.51±10.80	95.34±3.87
Fallow_smooth	95.23±0.63	94.16±8.27	97.94±2.14	95.19±4.88	95.40±7.26	97.94±2.43	95.45±5.73	95.51±5.13
Stubble	99.25±0.91	99.10±0.98	99.91±0.11	99.5 ±0.98	99.53±0.73	99.98±0.03	97.62±4.78	99.72±0.58
Celery	98.82±0.33	98.87±1.56	99.59±0.42	96.4 ±3.54	98.77±1.45	99.68±0.48	97.85±2.42	97.77±2.16
Grapes_untrained	78.50±0.57	83.95±9.71	72.24±37.12	88.51±6.99	90.81±4.35	79.91±19.12	92.34±4.01	90.80±2.96
Soil_vinyard_develop	94.11±0.50	99.98±0.04	99.85±0.18	99.32±0.67	99.67 ±0.72	99.54±0.58	95.88±6.80	97.58±1.30
Corn_senesced_green_weeds	85.56±0.36	93.96±1.35	93.54±8.37	98.22±1.50	93.29±6.93	95.65±4.92	89.66±12.38	94.66±12.40
Lettuce_romaine_4wk	90.63±0.78	90.53±12.29	90.14±11.32	98.54±2.18	89.70±16.64	93.52±6.89	89.31±16.39	96.20±4.50
Lettuce_romaine_5wk	99.48±0.03	92.17±10.92	95.51±5.28	96.43±2.46	87.34±24.58	97.40±2.65	94.70±5.59	94.68±9.16
Lettuce_romaine_6wk	20.08±2.47	100±0.00	99.18±0.98	93.21±6.55	64.61±43.87	98.65±1.55	87.77±13.04	83.90±29.24
Lettuce_romaine_7wk	66.29±1.67	91.85±9.60	98.81±1.76	85.59±19.38	91.92±10.08	99.25±1.50	84.53±29.87	97.45±2.72
Vinyard_untrained	59.14±1.06	83.68±4.81	89.74±12.88	79.57±11.79	83.72±13.10	78.71±38.01	80.70±16.29	78.38±15.95
Vinyard_vertical_trellis	66.96±0.78	91.59±10.84	44.59±30.50	78.98±18.37	44.24±32.06	51.82±32.59	76.08±27.38	69.21±39.33

TABLE V
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE PAVIA DATASET USING 200 TRAINING SAMPLES

Method	SVM	SSRN	CNN	MLP	PCA-CNN	MC-Dropout	**BDL-G^2**	**BGN-G^2 -BFL**
OA(%)	80.06±1.52	92.00±2.01	91.62±0.59	92.13±2.76	91.71±1.18	90.97±1.81	91.31±1.01	92.81±1.82
AA(%)	69.48±3.03	87.88±4.61	82.95±1.82	88.48±3.94	85.17±2.19	80.57±3.38	83.08±2.42	87.32±2.83
K×100	74.19±2.02	89.52±2.72	89.43±0.71	90.07±3.48	89.55±1.49	88.55±2.35	88.98±1.30	90.95±2.29
ECE (%)	23.07±2.84	23.23±5.62	9.06±2.98	5.93±0.61	6.27±1.21	7.76±2.45	5.86±1.57	5.20±1.66
Asphalt	88.75±2.70	94.99±3.82	91.64±4.03	92.68±4.40	94.75±1.61	95.70±1.04	92.90±3.44	90.91±4.56
Meadows	94.94±1.47	97.62±1.87	99.17±0.33	97.20±1.62	96.81±2.07	99.16±1.24	99.43±0.74	98.46 ±2.09
Gravel	34.89±15.46	61.45±12.11	46.16±18.32	79.84±6.59	61.90±16.72	45.16±25.57	61.47±23.60	74.41±15.96
Trees	64.11±9.76	96.34±2.68	90.66±4.27	94.53±1.02	94.30±3.40	86.90±10.70	80.37±9.01	86.74±7.83
Metal sheets	89.35±8.20	99.49±0.26	97.58±3.33	98.49±1.11	91.70±9.54	98.35±1.52	97.22±3.05	96.67±8.00
Bare soil	66.54±6.15	81.84±15.59	99.25±0.69	86.72±3.97	92.70±6.85	96.91±2.42	94.52±3.48	97.30±3.05
Bitumen	55.77±22.79	70.61±39.42	63.31±18.21	88.23±11.54	60.47±19.00	56.03±26.97	65.43±18.51	78.84±13.25
Bricks	73.39±7.31	91.06±1.70	85.97±7.89	88.25±6.61	90.58±4.98	85.32±10.31	89.58±8.70	90.72±4.94
Shadow	57.61±21.51	97.49±1.80	72.84±16.96	70.38±14.97	83.26±10.63	61.62±18.65	66.82±14.34	71.87±8.41

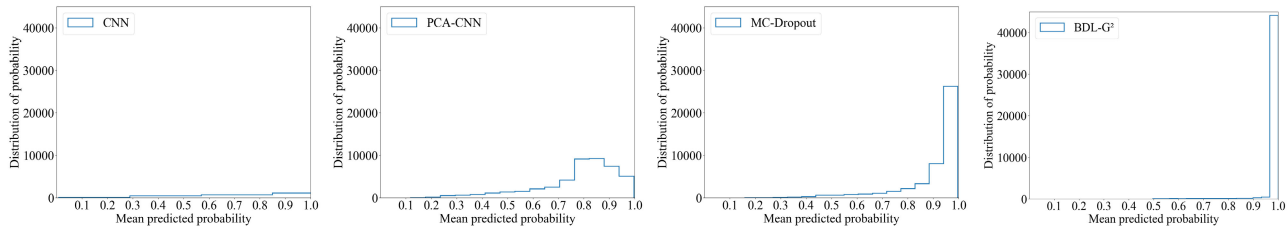


Fig. 6. Distribution of the output probabilities for different methods on the Salinas dataset. (a) CNN. (b) PCA-CNN. (c) MC-Dropout. (d) BDL-G^2.

The results demonstrate that the proposed BDL-G^2 is able to learn Gabor parameters with Gaussian distribution for HSI classification.

C. Experimental Results for BDL-G^2-BFL

1) Classification Performance of the Proposed BDL-G^2-BFL: Classification results of the proposed BDL-G^2-BFL are reported at the last column in Tables IV–VI. Obviously, the proposed BDL-G^2-BFL yields the highest accuracy on the three datasets. Compared with CNN, the proposed BDL-G^2-BFL has 2.94%, 1.19%, and 2.12% improvements on the Salinas, Pavia, and Indian Pines datasets, respectively. Moreover, the proposed BDL-G^2-BFL has the gains of 9.64%, 1.92%, 1.92%, and 1.65% over SVM, SSRN, MLP, and MC-Dropout on the Salinas dataset. In addition, in comparison of the six

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TABLE VI
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE INDIAN PINES DATASET USING 200 TRAINING SAMPLES

Method	SVM	SSRN	CNN	MLP	PCA-CNN	MC-Dropout	**BDL-G^2**	**BGN-G^2 -BFL**
OA(%)	79.75±1.89	82.37±6.45	86.16±2.12	87.18±1.53	87.70±1.37	86.13±1.15	87.74±1.54	88.28±1.25
AA(%)	60.11±2.56	67.44±7.68	65.79±6.58	72.27±4.54	66.51±5.18	66.04±5.73	72.24±6.57	73.00±5.26
K×100	75.82±2.35	79.87±7.32	83.35±2.55	84.65±1.85	85.23±1.68	83.31±1.42	85.29±1.84	85.94±1.53
ECE (%)	47.81±10.38	23.22±1.32	9.06±2.98	10.10±1.05	16.79±1.94	7.76±2.45	6.13±1.30	5.63±1.47
Alfalfa	38.67±26.66	16.60±36.48	50.82±32.72	60.69±40.45	42.18±26.04	47.14±29.41	59.96±40.06	53.45±36.59
Corn-notill	80.01±9.17	71.55±22.58	85.52±7.43	89.16±4.33	91.18±2.65	88.82±5.91	86.79±5.17	89.58±4.71
Corn-min	52.50±11.83	79.21±22.83	68.18±9.96	59.89±17.34	60.14±22.74	67.86±9.83	59.90±11.83	62.70±15.86
Corn	23.85±22.71	48.43±28.18	43.26±32.56	44.68±38.53	35.35±33.13	44.25±33.39	56.14±33.97	65.42±34.50
Grass-pasture	61.00±16.10	84.05±11.08	60.95±12.54	65.29±12.39	53.06±17.82	62.82±9.26	71.29±23.85	66.28±21.07
Grass-trees	91.76±4.49	94.17±8.41	91.54±6.50	93.30±3.13	92.09±5.31	91.53±9.76	94.58±2.62	94.58±3.15
Grass-pasture-mowed	16.58±23.12	26.28±42.86	42.61±24.59	21.28±39.84	31.00±35.98	46.61±31.47	53.85±45.05	57.31±47.25
Hay-windrowed	87.89±8.56	96.25±5.91	84.14±19.09	90.05±7.87	87.47±10.60	87.15±11.50	86.84±17.08	82.62±18.92
Oats	42.21±37.64	3.51±8.59	20.00±40.00	44.01±47.16	25.73±29.60	20.00±40.00	7.37±22.11	8.42 ±25.26
Soybean-notill	74.57±8.69	82.35±15.78	89.02±6.31	86.01±9.06	89.39±5.80	87.69±7.32	85.50±4.48	87.09±3.98
Soybean-mintill	92.74±3.23	84.04±11.11	95.86±2.49	91.82±2.53	94.88±1.70	94.34±2.48	94.32±3.91	94.07±2.31
Soybean-clean	48.60±13.61	75.61±19.87	48.14±20.02	65.56±17.23	66.60±21.51	44.24±12.99	69.07±24.03	63.62±18.96
Wheat	61.59±25.50	76.11±37.70	77.07±38.85	83.02±16.25	91.10±13.30	77.67±38.89	92.04±10.45	92.36±7.31
Woods	94.26±2.15	99.17±1.06	98.57±0.94	96.30±2.46	97.50±2.40	98.28±1.35	96.14±4.71	97.73±2.14
Buildings-Grass-Trees	52.78±28.23	74.95±17.30	51.22±34.29	84.59±23.84	64.48±29.23	47.75±34.26	72.29±32.81	79.81±26.47
Stone-Steel-Towers	42.78±34.42	66.79±45.21	45.78±22.17	66.36±30.98	41.97±30.03	50.46±20.01	69.86±17.55	73.05±18.78

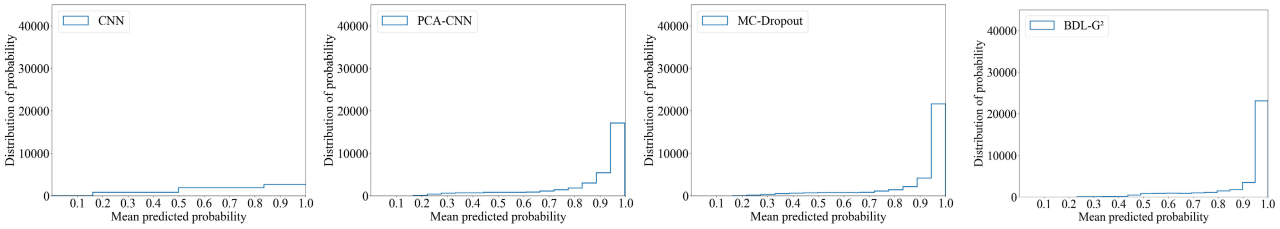


Fig. 7. Distribution of the output probabilities for different methods on the Pavia dataset. (a) CNN. (b) PCA-CNN. (c) MC-Dropout. (d) BDL-G^2.

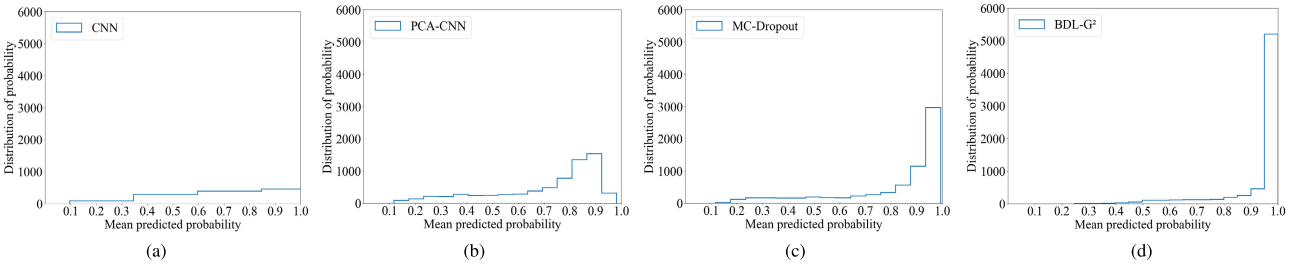


Fig. 8. Distribution of the output probabilities for different methods on the Indian Pines dataset. (a) CNN. (b) PCA-CNN. (c) MC-Dropout. (d) BDL-G^2.

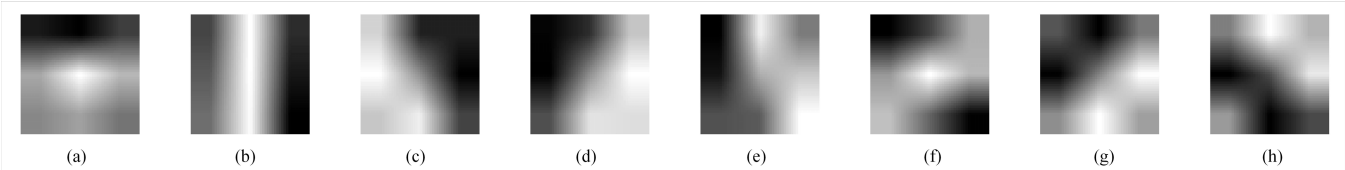


Fig. 9. Learned Gabor filters in the first Bayesian Gabor layer. (a) First learned Gabor filter, (b) second learned Gabor filter, (c) third learned Gabor filter, (d) fourth learned Gabor filter, (e) fifth learned Gabor filter, (f) sixth learned Gabor filter, (g) seventh learned Gabor filter, and (h) eighth learned Gabor filter.

compared methods (i.e., SVM, SSRN, CNN, MLP, PCA-CNN, and MC-Dropout), the proposed BDL-G^2-BFL has the lowest uncertainty in terms of ECE. The Pavia and Indian Pines datasets have similar results with the Salinas dataset. From the results, it can be concluded that the proposed BDL-G^2-BFL is able to enhance the classification performance and reduce the uncertainty for HSI classification.

2) Visualization of Calibration Curves: Figs. 11–13 show the calibration curves are used in this article to test the uncertainty of different models on the three datasets. This is a

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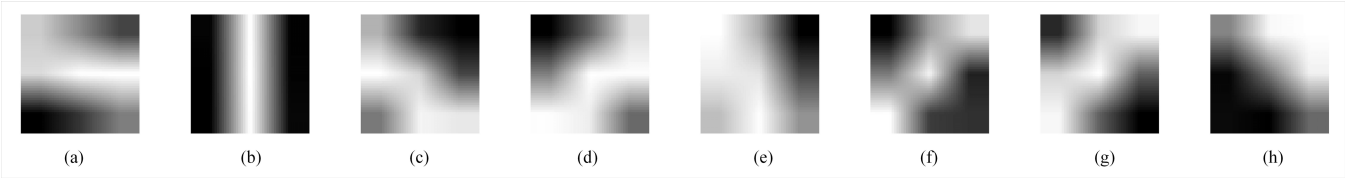


Fig. 10. Learned Gabor filters in the second Bayesian Gabor layer. (a) First learned Gabor filter, (b) second learned Gabor filter, (c) third learned Gabor filter, (d) fourth learned Gabor filter, (e) fifth learned Gabor filter, (f) sixth learned Gabor filter, (g) seventh learned Gabor filter, and (h) eighth learned Gabor filter.

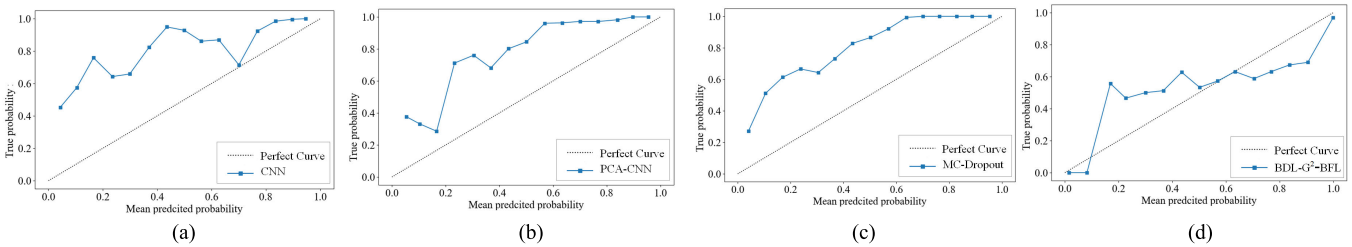


Fig. 11. Calibration curves of different methods on the Salinas dataset. (a) CNN. (b) PCA-CNN. (c) MC-Dropout. (d) BDL-G^2-BFL.

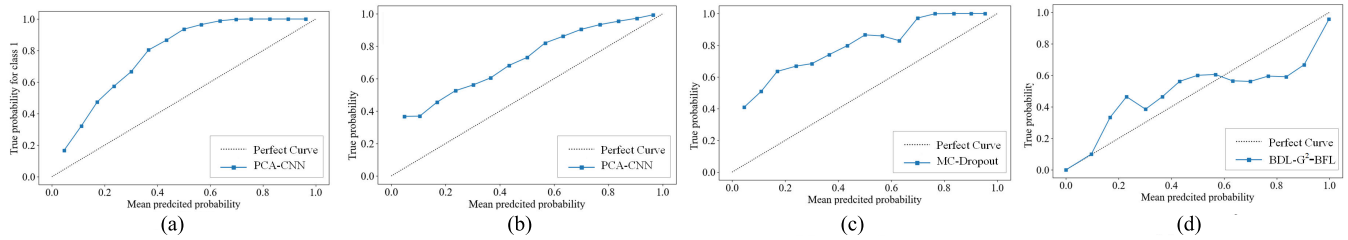


Fig. 12. Calibration curves of different methods on the Pavia dataset. (a) CNN. (b) PCA-CNN. (c) MC-Dropout. (d) BDL-G^2-BFL.

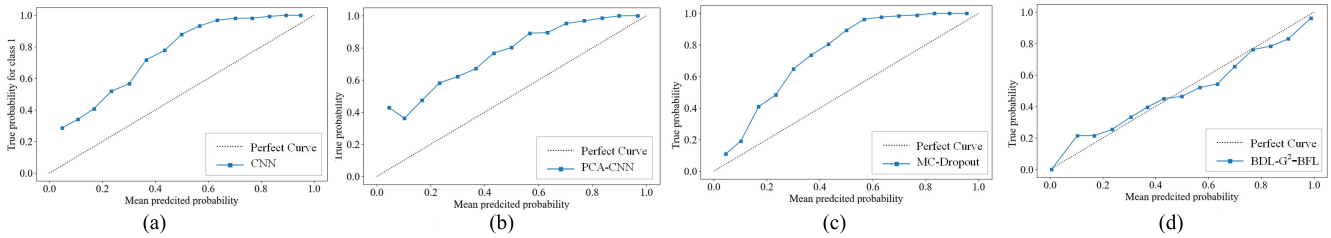


Fig. 13. Calibration curves of different methods on the Indian Pines dataset. (a) CNN. (b) PCA-CNN. (c) MC-Dropout. (d) BDL-G^2-BFL.

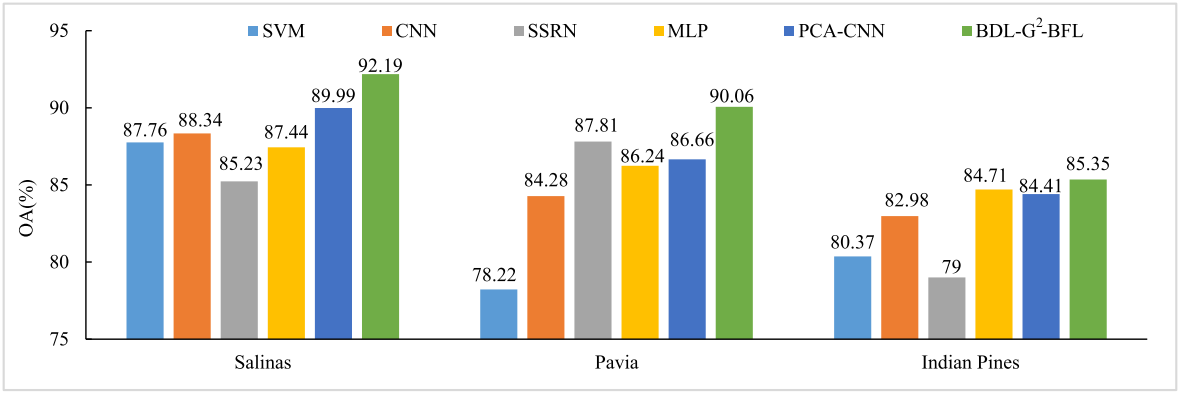


Fig. 14. Test accuracy (%) comparisons under different methods on the two datasets with 150 training samples.

curve with the predicted probability as the abscissa and the true label as the ordinate. It is hoped that the output probabilities of the model and the true value are as close as possible, preferably the two are equal, so the ideal case is that the probability calibration curve of the model is to the diagonal, which indicates that the uncertainty of the model is the lowest.

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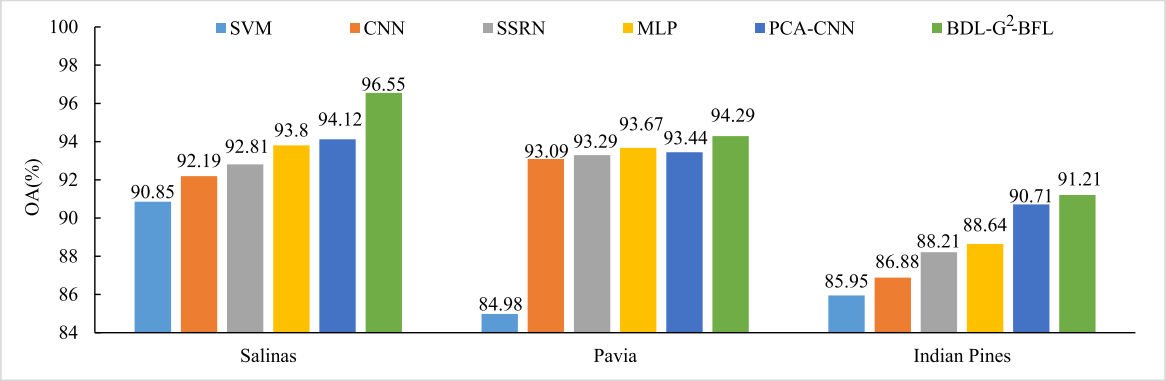


Fig. 15. Test accuracy (%) comparisons under different methods on the two datasets with 300 training samples.

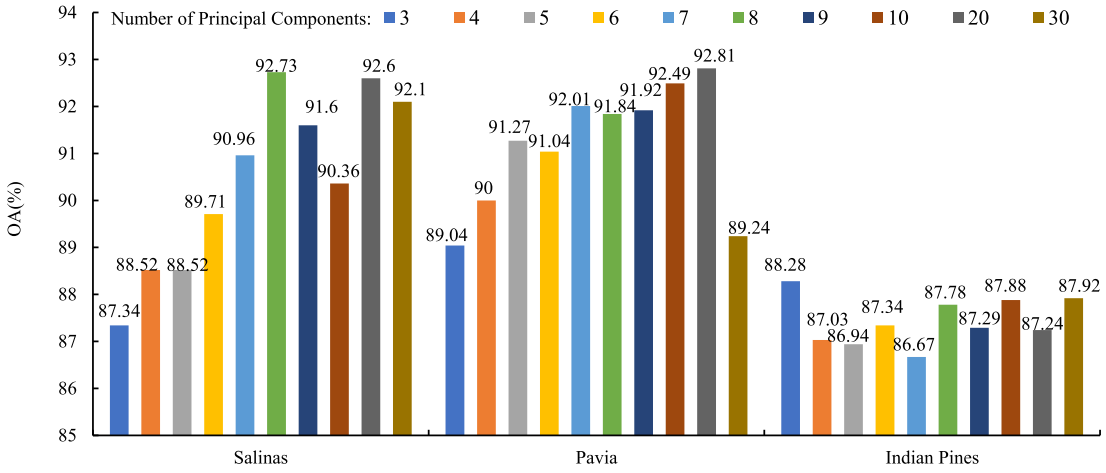


Fig. 16. Comparison of different numbers of principal components on the three datasets.

As can be seen, Figs. 11–13(a)–(d) represent the calibration curves of CNN, PCA-CNN, MC-Dropout, and the proposed BDL-G^2-BFL on the Salinas, Pavia, and Indian Pines datasets, respectively. Take the Indian Pines dataset as an example, in Fig. 11(e), it can be seen that the proposed BDL-G^2-BFL is closest to diagonal, compared with other models. The visual results demonstrate that the proposed BDL-G^2-BFL is effective by reducing the uncertainty.

3) Different Number of Training Samples: Figs. 14 and 15 show the OA of different methods (i.e., SVM, CNN, SSRN, MLP, PCA-CNN, and BDL-G^2-BFL) on the three HSI datasets with 150 and 300 training samples, respectively. Specifically, the proposed BDL-G^2-BFL achieves the best overall classification accuracies on the three datasets with 150 and 300 training samples. On the Pavia dataset, when the number of training samples is 150, the proposed BDL-G^2-BFL improves the classification accuracy by 11.84%, 5.78%, 2.25%, 3.82%, and 3.4% in comparison of SVM, CNN, SSRN, MLP, and PCA-CNN, respectively. In addition, the proposed BDL-G^2-BFL reaches 96.55%, 94.29%, and 91.21% in terms of OA on the three datasets, respectively. All the results demonstrate that the proposed methods are effective with different training samples.

D. Ablation Study

1) Ablation Study on the Number of Principal Components: In the experiments, PCA is applied to the proposed approaches to extract spectral information for HSI classification. Fig. 16 shows the performance of different numbers of principal components on the Salinas, Pavia, and Indian Pines, respectively. As can be seen, OA tends to grow on the Salinas dataset, and when the number of principal components is eight, the accuracy reaches the highest. Then OA decreases as the number of principal components is growing. While for the Pavia dataset, when the number of principal components is 20, the accuracy is the highest. For the Indian Pines dataset, when the number of principal components is three, the satisfactory accuracy is obtained. Therefore, the number of principal components is set to 8, 20, and 3 on the Salinas, Pavia, and Indian Pines, respectively.

2) Ablation Study on Bayesian Gabor Layer and Bayesian FC Layer: In this article, the uncertainty measure is investigated at both the FE and classification stages. At the FE stage, the Gabor layer is without uncertainty measure, while the Bayesian Gabor layer is with uncertainty measure. Similarly, at the classification stage, the FC layer is without uncertainty measure, while the Bayesian FC layer is with uncertainty measure.

To analyze the contributions of the Bayesian Gabor layer and Bayesian FC layer for HSI classification, ablation studies are carried out in the experiments. Tables VII–IX report the classification performance in terms of OA, AA, K, and ECE. As can be seen, the proposed BDL-G^2-BFL, which measures

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TABLE VII
ABLATION EXPERIMENTS ON THE SALINAS DATASET

Feature Extraction		Classification		OA (%)	AA (%)	K×100	ECE (%)
Gabor Layer	Bayesian Gabor Layer	FC Layer	Bayesian FC Layer				
✓	×	✓	×	88.49±6.81	85.99±7.91	87.13±7.55	6.72±4.57
✓	×	×	✓	91.88±1.36	90.02±3.57	90.86±1.54	6.23±2.11
×	✓	✓	×	91.97±3.62	90.72±4.41	90.96±4.13	5.83±3.24
×	✓	×	✓	92.73±1.36	92.39±2.93	91.84±1.54	3.93±1.54

TABLE VIII
ABLATION EXPERIMENTS ON THE PAVIA DATASET

Feature Extraction		Classification		OA (%)	AA (%)	K×100	ECE (%)
Gabor Layer	Bayesian Gabor Layer	FC Layer	Bayesian FC Layer				
✓	×	✓	×	87.62±0.98	77.07±3.86	84.38±1.31	6.74±0.42
✓	×	×	✓	90.84±1.42	84.26±1.98	88.43±1.79	6.48±1.57
×	✓	✓	×	91.31±1.01	83.08±2.42	88.98±1.30	5.86±1.57
×	✓	×	✓	92.81±1.82	87.32±2.83	90.95±2.29	5.20±1.66

TABLE IX
ABLATION EXPERIMENTS ON THE INDIAN PINES DATASET

Feature Extraction		Classification		OA (%)	AA (%)	K×100	ECE (%)
Gabor Layer	Bayesian Gabor Layer	FC Layer	Bayesian FC Layer				
✓	×	✓	×	83.39±1.64	69.24±6.25	80.07±2.05	8.32±1.45
✓	×	×	✓	84.93±1.24	66.95±4.27	81.89±1.50	7.21±3.41
×	✓	✓	×	87.74±1.54	72.24±6.57	85.29±1.84	6.13±1.30
×	✓	×	✓	88.28±1.25	73.00±5.26	85.94±1.53	5.63±1.47

uncertainty at both the Bayesian Gabor layer and Bayesian FC layer, obtains the best classification results in terms of OA, AA, and K. Moreover, the uncertainty of the proposed BDL-G^2 is the lowest in terms of ECE which is below 6% on all the datasets. The results reflect that BDL-G^2-BFL is conducive to the improvement of accuracy and the reduction of uncertainty.

E. Comparison of Complexity and Time

Table X reports the training time, test time, and total parameters of different methods (i.e., SVM, CNN, SSRN, MLP, PCA-CNN, MC-Dropout, BDL-G^2, and BDL-G^2-BFL) on the three datasets. The experiments are implemented on a computer with an Intel Core i7-11700 processor with 2.5 GHz and an NVIDIA GeForce RTX 3070 Ti graphical processing unit (GPU). It can be seen that compared with other methods, the complexity (i.e., number of total parameters) of the proposed BDL-G^2 and BDL-G^2-BFL is the lowest. Besides, compared with SSRN and MLP, the proposed BDL-G^2 and BDL-G^2-BFL reduce the training and test time on the three datasets. Since the proposed methods need more epochs to train the model, the training time is longer than CNN and PCA-CNN. But the proposed methods achieve higher classification accuracy, which demonstrated the proposed methods are relatively effective.

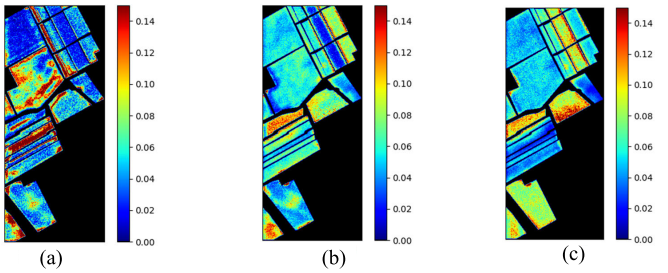


Fig. 17. Uncertainty map of different methods on the Salinas dataset. (a) MC-Dropout. (b) BDL-G^2. (c) BDL-G^2-BFL.

F. Uncertainty Map

To estimate the performance of uncertainty and analyze what they capture in the proposed BDL-G^2 and BDL-G^2-BFL, Figs. 17–19 display the uncertainty maps of different methods (i.e., (a) MC-Dropout, (b) BDL-G^2, and (c) BDL-G^2-BFL) by doing several forward passes through the model with the

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TABLE X
TIME CONSUMPTION AND COMPUTATIONAL COMPLEXITY ON THE HSI DATASETS

Datasets	Methods	Training Time (min.)	Test Time (min.)	Total parameters (M)
Salinas	SVM	13.25	452.6	-
	CNN	0.48	1.38	0.13
	SSRN	2.36	1.11	0.24
	MLP	1.07	1.05	0.72
	PCA-CNN	0.02	0.14	0.12
	MC-Dropout	1.64	1.21	0.14
	BDL-G^2	0.93	0.22	0.06
	BDL-G^2-BFL	0.94	0.22	0.06
Pavia	SVM	12.67	187.15	-
	CNN	0.31	0.48	0.13
	SSRN	1.28	0.34	0.14
	MLP	1.07	1.05	0.72
	PCA-CNN	0.02	0.11	0.12
	MC-Dropout	0.58	0.59	0.13
	BDL-G^2	0.89	0.23	0.06
	BDL-G^2-BFL	0.9	0.23	0.06
Indian Pines	SVM	19.73	218.9	-
	CNN	0.39	0.15	0.13
	SSRN	2.21	0.06	0.23
	MLP	2.1	0.09	0.91
	PCA-CNN	0.01	0.02	0.12
	MC-Dropout	0.53	0.17	0.13
	BDL-G^2	0.8	0.04	0.06
	BDL-G^2-BFL	0.81	0.04	0.06

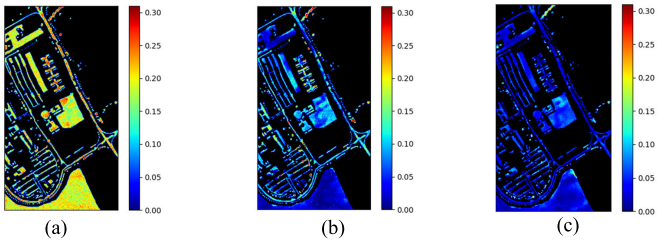


Fig. 18. Uncertainty map of different methods on the Pavia dataset. (a) MC-Dropout. (b) BDL-G^2. (c) BDL-G^2-BFL.

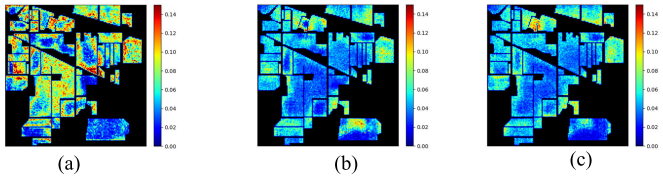


Fig. 19. Uncertainty map of different methods on the Indian Pines dataset. (a) MC-Dropout. (b) BDL-G^2. (c) BDL-G^2-BFL.

same input data. Therefore, the uncertainty map is obtained by computing the variance of the distribution of the outputs.

In Figs. 17–19, color bar is made to give a better visualization, and the brighter area represents the high uncertainty and the black area is the unlabeled samples without considering the uncertainty. It can be observed that the areas of uncertainty maps produced by BDL-G^2-BFL are darker than the maps produced by MC-Dropout and BDL-G^2. Specifically, in the upper left corner area of Fig. 17(a), this area is brighter compared with the areas in Fig. 17(b) and (c), which means the MC-Dropout brings high uncertainty. Similarly, in the center

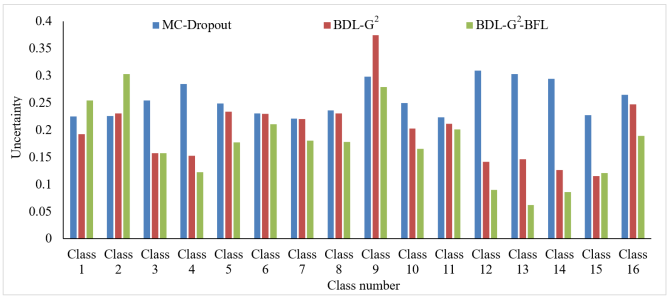


Fig. 20. Uncertainty values of each class on the Salinas dataset.

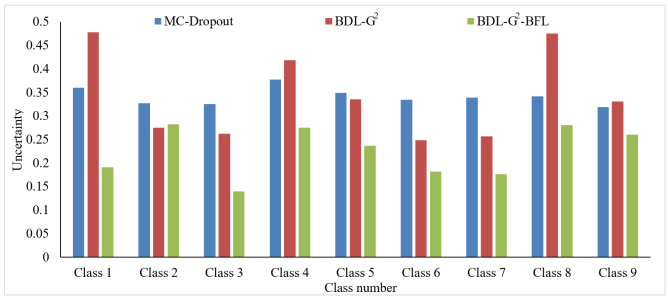


Fig. 21. Uncertainty values of each class on the Pavia dataset.

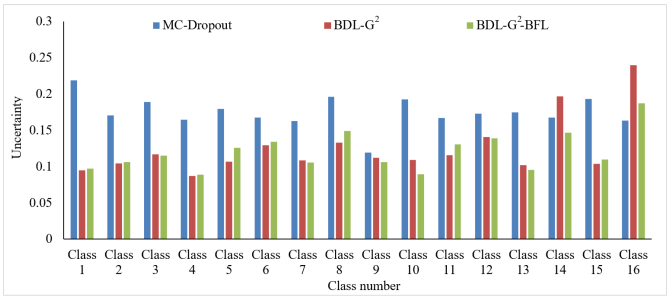


Fig. 22. Uncertainty values of each class on the Indian Pines dataset.

of Figs. 18 and 19, the proposed BDL-G^2 and BDL-G^2-BFL obtain darker areas compared with MC-Dropout, which means the proposed methods have low uncertainty on the Pavia and Indian Pines datasets. The above visual results demonstrate the proposed models have low uncertainty about its results.

Furthermore, to give a quantitative comparison, the uncertainty values of each class in the uncertainty maps obtained by different methods (i.e., MC-Dropout, BDL-G^2, and BDL-G^2-BFL) are listed in Figs. 20–22. It can be seen that the uncertainty values of different classes are various. Especially, in Fig. 21, uncertainty values of all the classes of BDL-G^2-BFL, which are highlighted in green, are the lowest ones. In addition, compared with MC-Dropout and BDL-G^2, most of the classes of the proposed BDL-G^2-BFL have the lowest uncertainty values. The above results demonstrate that the proposed methods have low uncertainty.

G. Classification Map

The qualitative maps of various models including CNN, PCA-CNN, SSRN, MC-Dropout, BDL-G^2, and BDL-G^2-BFL on the Salinas, Pavia, and Indian Pines datasets are

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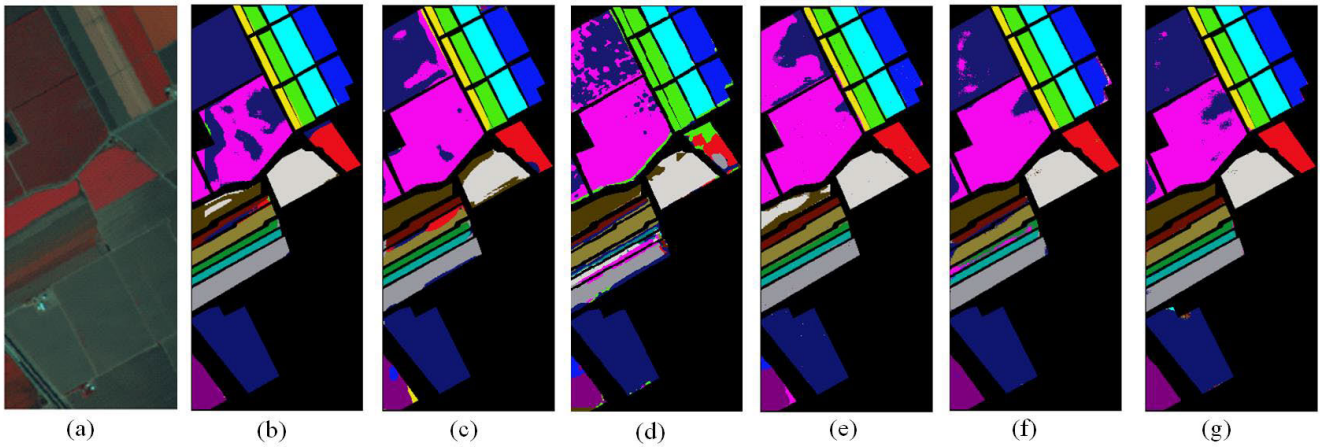


Fig. 23. Salinas. (a) False-color composite image. The classification maps using (b) CNN, (c) PCA-CNN, (d) SSRN, (e) MC-Dropout, (f) BDL-G^2, and (g) BDL-G^2-BFL.

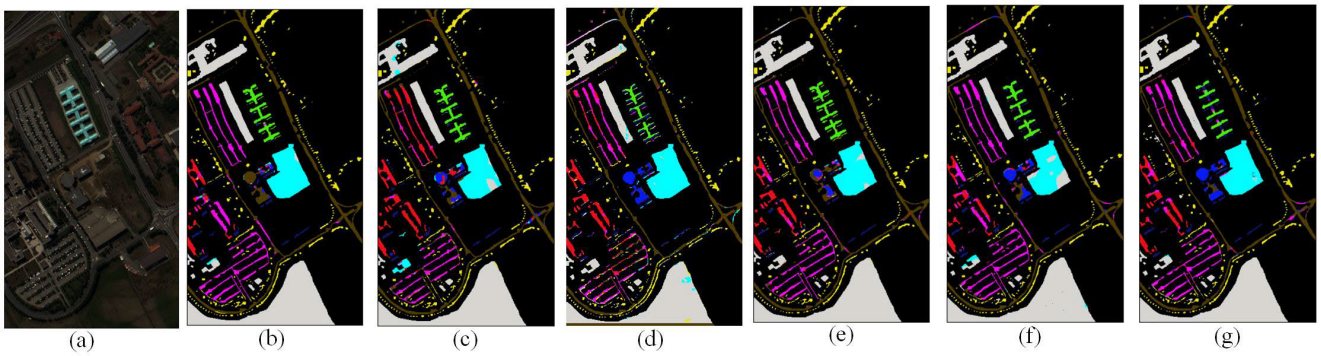


Fig. 24. Pavia. (a) False-color composite image. The classification maps using (b) CNN, (c) PCA-CNN, (d) SSRN, (e) MC-Dropout, (f) BDL-G^2, and (g) BDL-G^2-BFL.

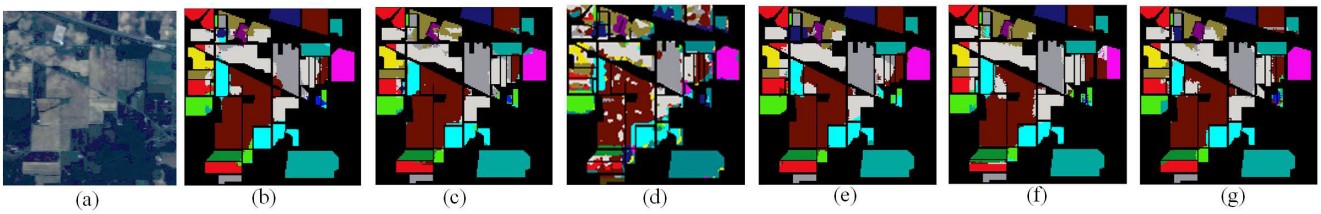


Fig. 25. Indian Pines. (a) False-color composite image. The classification maps using (b) CNN, (c) PCA-CNN, (d) SSRN, (e) MC-Dropout, (f) BDL-G^2, and (g) BDL-G^2-BFL.

provided in Figs. 23–25, respectively. It can be seen that the proposed BDL-G^2-BFL acquires smoother maps. Considering the Salinas dataset shown in Fig. 23, the border pixels between adjacent ground objects can be separated discriminatively in Fig. 23(g). With respect to the Pavia dataset, there exists amount of misclassification in the center of Fig. 24(a)–(e), while the proposed BDL-G^2-BFL in Fig. 24(g) is more robust for border pixel classification. Meanwhile, in Fig. 25, compared with other models (i.e, CNN, PCA-CNN, SSRN, and MC-Dropout), there are less noisy pixels on the area of soybean-notill (gray color) on the Indian Pines in Fig. 25(g). The visual results demonstrate the proposed methods (i.e, BDL-G^2 and BDL-G^2-BFL) have better classification performance.

V. Conclusion

In this article, to establish an accurate HSI classification with low uncertainty, two types of frameworks (i.e., BDL-G^2 and BDL-G^2-BFL) based on Bayesian inference were proposed to reduce the model uncertainty for HSI classification. First, the proposed BDL-G^2 modeled the uncertainty at the FE stage, which investigated two Bayesian Gabor layers by learning probability distributions of all the Gabor parameters. The proposed BDL-G^2 enhanced the structure characteristic of HSI using the Gabor filtering and provided uncertainty estimation by exploring two Bayesian Gabor layers. Second, to further estimate the model uncertainty at the classification stage, the proposed BDL-G^2-BFL was proposed which combined with BDL-G^2. In BDL-G^2-BFL, we explored a Bayesian FC layer with a model-free distribution on the parameters of FC layer, which provided model uncertainty by adjusting parameters in distribution adaptively. By constructing model uncertainty at both the feature stage and classification stage, the proposed methods enhanced the accuracy and reduced the model uncertainty simultaneously.

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